

INGOBOKA

Ubwishingizi bukugoboka mu bihe bikomeye.

Enterprise Digital Microinsurance, Policy Wallet and Claims Platform



System Concept, Problem Definition, Solution Scope, Impact and Enterprise Architecture

Prepared for: UMURUNGI Olga
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Document purpose

This editable concept document defines the proposed product, the problem it addresses, the services included, the expected impact, and a Spring Boot enterprise architecture. It separates the six-day hackathon MVP from the long-term production platform so the team can deliver a credible prototype without overpromising.

Working name subject to Kinyarwanda language review, trademark screening and regulatory validation.

Document Control

Item	Detail
Document title	Ingoboka - Enterprise Digital Microinsurance, Policy Wallet and Claims Platform
Version	1.0 - Concept and architecture draft
Date	15 June 2026
Team Lead	Mahinga Rodin & Ruyange Arnold
Primary audience	Hackathon reviewers, product team, designers, developers, potential insurer partners and mentors
Status	Editable working document
Technology direction	Spring Boot backend, PostgreSQL, responsive React/Next.js PWA, secure REST APIs
Scope note	The MVP demonstrates a digital distribution and claims workflow; it does not operate as an insurer or make autonomous underwriting decisions.

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Editing note

After making changes in Microsoft Word or Google Docs, update any page numbering or contents list if the document length changes.

1. Executive Summary

Ingoboka is a proposed Rwanda-focused digital platform that makes inclusive insurance easier to understand, access, manage and claim. The platform is designed primarily for citizens whose incomes are irregular or whose access to conventional insurance channels is limited, including informal workers, market vendors, moto riders, farmers, cooperative members, young adults and micro-business owners.

The platform will not replace licensed insurers. It will provide a secure B2B2C technology layer through which approved insurance partners can publish suitable products, enroll customers, collect or orchestrate flexible premium contributions, issue digital policy records, support claims, communicate decisions and monitor performance. Final product pricing, underwriting and claim decisions remain the responsibility of authorized insurance providers.

The hackathon MVP will demonstrate one complete journey: a citizen registers, gives informed consent, completes a short needs assessment, selects an approved demo product, makes a sandbox payment, receives a digital policy card with a privacy-safe QR code, submits a claim and follows the claim through review to decision. An insurer portal will show product management, policy activity, claim review and basic analytics.

For delivery within six days, the backend should be implemented as a modular Spring Boot application with clear domain boundaries. This provides enterprise-quality structure while avoiding the deployment and operational overhead of premature microservices. The modules can later be separated into independently deployable services as transaction volumes, partnerships and team size grow.

One-sentence value proposition

Ingoboka helps licensed insurers reach underserved Rwandans through understandable, flexible and transparent digital insurance journeys, while giving citizens a secure policy wallet and a clear path from enrollment to claim resolution.

2. Product Name and Brand Direction

2.1 Recommended name: Ingoboka

Ingoboka is recommended as the working Kinyarwanda brand name because it communicates the idea of protection and prevention. It is short, memorable and suitable for a digital product. To remove any ambiguity, the name should always be accompanied by a clear insurance descriptor during the hackathon and early market testing.

Brand element	Recommended wording
Primary name	Ingoboka
Kinyarwanda descriptor	Urubuga rw'ubwishingizi bworoshye kandi bugera kuri bose
English descriptor	Digital Microinsurance, Policy Wallet and Claims Platform
Tagline	Ubwishingizi bworoshye, bwumvikana kandi bugera kuri bose.
Brand promise	Protection that citizens can understand, access and trust.

Validation required

Before public launch, validate the name with native-language users from different districts, conduct a trademark and company-name search, and confirm that the brand does not imply that the platform itself is a licensed insurer.

3. Background and Opportunity

The Fintech Innovation Hackathon 2026 seeks practical technology-driven solutions that advance financial inclusion, interoperability, efficiency and responsible digital financial services. One of the official thematic areas is enhancing access and innovation in digital insurance. The registration form also distinguishes between an idea, an early prototype and a functional MVP, so the project must show a working end-to-end flow rather than design screens alone. [1]

Rwanda has achieved broad financial inclusion, but meaningful use of insurance remains limited and uneven. The National Financial Inclusion Roadmap 2026-2030 reports more than 600,000 policyholders, approximately 7.5% of the adult population, with substantial concentration in urban districts. It sets a target of 15% insured adults and calls for product innovation, digital insurance, broader distribution channels, affordable microinsurance and mobile or fintech-enabled claims processing. [2]

This creates a strong opportunity for a localized digital platform that does not attempt to become an insurer, but instead reduces the access, understanding, distribution and service barriers between citizens and licensed providers.

Observed opportunity	How Ingoboka responds
Low and uneven insurance reach	Mobile-first enrollment, assisted agent mode, multilingual journeys and configurable low-value products.
Products may be difficult to understand	Plain-language coverage, exclusions, waiting periods and claim requirements, shown before consent.
Irregular income does not fit rigid payment cycles	Configurable daily, weekly or monthly contribution schedules where permitted by the insurer.
Paper or fragmented policy records	Secure digital policy wallet, downloadable certificate and limited QR verification.
Unclear claims progress	Step-by-step claim timeline, missing-document requests, decision reasons and appeal workflow.
Insurers need efficient digital distribution	Partner portal, APIs, product configuration, reconciliation, reporting and audit trails.

4. Problem Definition

4.1 Core problem statement

Many Rwandans who are economically active remain insufficiently protected against accidents, illness-related income shocks, funeral costs, loss of small-business assets and other emergencies. The challenge is not only whether insurance products exist. It is also whether people can discover an appropriate product, understand its conditions, afford the payment pattern, retain proof of cover and complete a claim without confusion or unnecessary delay.

Informal workers and micro-entrepreneurs often receive income in small and irregular amounts. Conventional monthly or annual payment schedules may not match their cash flow. Product documents may use technical terms, and users may not clearly understand exclusions, waiting periods, required documents or the difference between enrollment and active cover. After an incident, the customer may not know where to submit evidence, what is missing, who is reviewing the claim or why a decision was made.

Insurers and intermediaries also face high costs when distribution, onboarding, document collection, communication and claims follow-up are handled manually or through disconnected tools. This limits the commercial viability of serving customers with low-value premiums and increases the risk of poor customer experience, mistrust and incomplete records.

4.2 Root causes

- Distribution channels do not always reach citizens where they work and transact.
- Insurance information is not consistently presented in simple Kinyarwanda and low-literacy formats.
- Premium schedules may not reflect daily or seasonal income patterns.

- Policy documents and proof of cover can be lost, inaccessible or difficult to verify.
- Claims often require multiple documents and repeated communication.
- Citizens may not receive transparent claim status updates or decision explanations.
- Insurer data, payment records, claim evidence and communication may be held in separate systems.
- Digital services can exclude users with low-end devices, limited data or low digital confidence.

4.3 Consequences

For citizens	For insurers and partners	For the financial system
Unexpected events consume savings, create debt or force sale of productive assets.	High acquisition and servicing cost for low-premium products.	Insurance growth remains concentrated and less inclusive.
Confusion about cover reduces trust and renewal.	Manual claims and reconciliation increase delay and error risk.	Limited usage weakens household and MSME financial resilience.
Missing information can cause abandoned or delayed claims.	Fragmented records reduce operational visibility.	Innovation is harder to evaluate without structured data and auditability.

5. Proposed Solution

5.1 Solution description

Ingoboka will be a multilingual, mobile-first insurance access and servicing platform connecting citizens, licensed insurers, approved agents and verification partners. It will provide one coherent journey from product discovery to policy management and claim resolution.

The citizen experience will prioritize simple language, visible consent, clear prices and benefits, low-data interaction, accessible navigation and transparent status tracking. The insurer experience will prioritize configurable products, workflow control, role-based access, complete audit trails, operational reporting and integration readiness.

The solution will use Spring Boot for the enterprise backend and expose versioned REST APIs. The hackathon version will be a modular monolith so that every business capability has a clear boundary but can be built, tested and deployed as one application. The long-term platform may separate high-volume capabilities such as payments, claims, notifications and reporting into independent services.

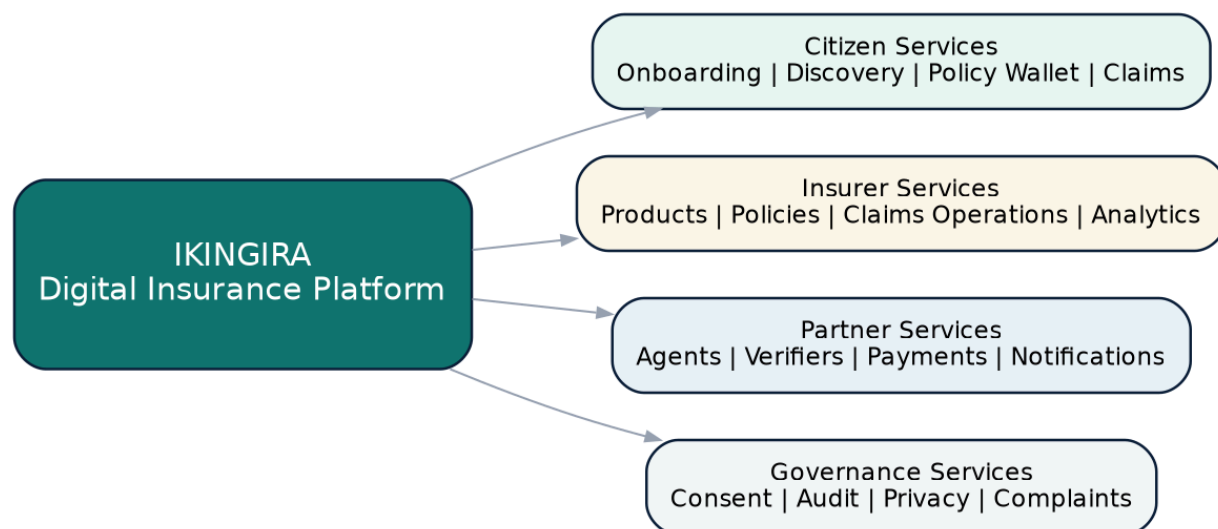


Figure 1. Ingoboka business service landscape

5.2 Core design principles

- **Understandable before digital:** A fast app is not inclusive if users do not understand what they are buying. Coverage, exclusions, waiting periods, payment obligations and claim requirements must be presented before consent.
- **Partner-led insurance:** Licensed insurers own product approval, pricing, underwriting and final claim decisions. Ingoboka provides technology, workflow and distribution support.
- **Privacy by design:** Collect only required data, separate sensitive documents, use role-based access, log access and provide consent and data-rights workflows.
- **Mobile and low-data first:** Optimize for affordable Android devices, limited bandwidth and assisted use. The PWA should work well without an app-store installation.
- **Transparent claims:** Every claim has a visible timeline, document checklist, reasoned decision and appeal path.
- **Configurable, not hard-coded:** Products, contribution cycles, required documents, eligibility rules and notifications should be data-driven.
- **Enterprise-ready but MVP-disciplined:** Use clean architecture, testing and security controls while limiting the six-day build to a complete demonstrable journey.

6. Target Users and Stakeholders

Stakeholder	Primary needs	Ingoboka capability
Citizen / policyholder	Understand products, enroll, pay, retain proof, claim and receive support.	PWA, policy wallet, claim wizard, timeline, notifications and complaints.
Beneficiary	Confirm benefit eligibility and provide required documents.	Controlled beneficiary access and assisted support workflow.
Insurance product manager	Configure products, rules, prices, documents and target segments.	Product catalogue, versioning, approvals and publication controls.
Claims officer	Receive complete claims, review evidence, request information and record decisions.	Claim queue, document viewer, notes, status controls and service-level timers.
Insurance operations manager	Monitor policies, payments, claims and team performance.	Dashboards, reports, alerts and audit trail.
Field agent / cooperative officer	Assist citizens with onboarding and claims without gaining excessive data access.	Assisted mode, delegated permissions and agent activity logging.
Customer-support officer	Resolve questions, complaints and appeals.	Case management, communication history and escalation.
Payment and notification partner	Exchange secure, reliable transaction or message events.	Integration APIs, callbacks, idempotency and reconciliation.
Platform administrator	Manage tenants, users, security settings and system configuration.	Tenant administration, RBAC, audit, monitoring and support tools.
Auditor / regulator-facing reviewer	Review evidence of controls and activity without changing operational data.	Read-only reports, immutable logs and export controls.

7. Services Covered by the Platform

The following catalogue defines the enterprise business services. In the six-day MVP, only the services marked Core MVP must be fully functional; the others may be represented by configuration, stubs or documented interfaces.

ID	Service	Scope	Hackathon level
S01	Identity and Access	Registration, login, password reset, JWT/OIDC-ready security, MFA for privileged users, roles and permissions.	Core MVP

ID	Service	Scope	Hackathon level
S02	Customer Profile and Consent	Personal profile, language, location, occupation, consent history, privacy notices and data-rights requests.	Core MVP
S03	KYC and Eligibility	Identity verification abstraction, duplicate detection, configurable eligibility checks and manual review.	MVP sandbox
S04	Needs Assessment and Education	Short questionnaire, risk needs, plain-language education, affordability warning and product guidance.	Core MVP
S05	Insurance Product Catalogue	Partner products, benefits, exclusions, waiting periods, premiums, schedules, documents, versions and publication.	Core MVP
S06	Enrollment and Policy Lifecycle	Application, consent, beneficiary, issuance, activation, renewal, grace period, lapse, cancellation and expiry.	Core MVP
S07	Premium and Payment Orchestration	Payment request, sandbox mobile-money flow, webhook handling, idempotency, receipts and reconciliation.	Core MVP sandbox
S08	Digital Policy Wallet and QR	Policy card, certificate download, privacy-safe QR token, status verification and expiry.	Core MVP
S09	Claims Management	First notice of loss, dynamic forms, document checklist, review, information request, decision, payout status and closure.	Core MVP
S10	Appeals, Complaints and Support	Reasoned decision, appeal submission, customer complaint, assignment, escalation and response history.	MVP basic
S11	Document Management	Secure upload, virus scan hook, classification, signed access links, retention and deletion controls.	Core MVP basic
S12	Notifications and Communication	In-app, SMS/email abstraction, templates, language selection, delivery status and communication log.	MVP basic
S13	Insurer and Partner Management	Organization onboarding, products, users, branches, agents, agreements and API credentials.	MVP basic
S14	Reporting and Analytics	Enrollment funnel, active policies, collections, claims, turnaround time, geography and segment indicators.	Core MVP basic
S15	Audit, Privacy and Compliance	Immutable business-event log, personal-data access log, consent evidence, retention and export.	Core MVP
S16	Integration Management	Payment, SMS, identity, insurer core system and future eKash/API adapters.	Interface-first
S17	Configuration and Localization	Kinyarwanda/English content, configurable statuses, rules, reference data and feature flags.	Core MVP
S18	Fraud and Risk Signals	Duplicate claims, unusual activity, device or payment signals and manual review flags.	Future

7.1 Important service boundary

Ingoboka is not the risk carrier

The platform may recommend or present products based on approved rules, but it must not describe itself as an insurer. It must not independently set premiums, underwrite risk, guarantee claims or replace the decision authority of a licensed insurer. Any automated assistance must remain explainable and subject to partner rules and human oversight.

8. Insurance Product Categories

The enterprise platform should support multiple configurable insurance categories. The hackathon MVP should implement one primary demo product deeply instead of implementing many products superficially.

Category	Example need	Recommended stage
Personal accident microinsurance	Accidental injury, temporary disability or accidental death benefit for moto riders, workers and traders.	Primary MVP demo product
Hospital cash / health support	Fixed daily benefit or approved support during hospitalization; not a medical diagnosis service.	Optional MVP bundle
Funeral and family protection	Defined benefit after an eligible death to reduce sudden family financial pressure.	Post-MVP
Micro-business asset cover	Fire, theft or defined damage affecting small stock, equipment or premises.	Post-MVP
Agriculture and livestock cover	Crop, weather or livestock risks for farmers and cooperatives.	Future partner-led module
Device or mobility-related cover	Defined loss or damage of approved devices or productive transport assets.	Future

MVP product recommendation

Demonstrate a partner-approved personal accident and hospital-cash product with daily, weekly and monthly contribution options. Use clearly labeled demo pricing and sandbox claim decisions.

9. End-to-End User Journeys



Figure 2. Core citizen journey demonstrated by the MVP

9.1 Citizen enrollment and policy journey

1. The citizen creates an account using a phone number or email and selects Kinyarwanda or English.
2. The platform presents a concise privacy notice and captures specific consent.
3. The citizen completes a needs and affordability assessment.
4. The platform displays only approved products and explains benefits, exclusions, waiting periods and claim documents.

5. The citizen selects a product, beneficiary and contribution schedule.
6. The payment service creates a sandbox payment request and records a successful callback.
7. The policy service activates the demo policy and generates a digital card and verification token.
8. The citizen receives a receipt, policy summary and renewal reminder.

9.2 Claim journey

1. The policyholder selects an active policy and starts a claim.
2. The claim form changes according to claim type and displays the required evidence.
3. Documents are uploaded securely and the platform confirms completeness.
4. The claim is submitted and receives a unique reference and service-level target.
5. A claims officer reviews the file, requests more information if needed and records notes.
6. The customer receives status updates without seeing internal confidential notes.
7. The insurer records an approved or rejected decision with a clear reason.
8. For an approved demo claim, payout status is simulated. For a rejected claim, the citizen can submit an appeal or complaint.

9.3 Insurer operations journey

1. A product manager creates or updates a draft product and submits it for internal approval.
2. Approved product versions are published to selected customer segments and channels.
3. Operations users monitor enrollment, payment and policy activation.
4. Claims officers work from prioritized queues and complete document and decision checklists.
5. Managers review turnaround times, exceptions, complaints and audit evidence.
6. Integration users reconcile payment and notification failures.

10. Functional Scope and Requirements

10.1 Role-based access model

Role	Key permissions	Restricted actions
Citizen	Own profile, consent, applications, policies, payments, claims, appeals and notifications.	Cannot view other users or change insurer decisions.
Agent	Create assisted sessions, support enrollment, upload authorized documents and view assigned cases.	Cannot approve claims or export broad customer data.
Claims officer	View assigned claims, request information, recommend or record authorized decisions.	Cannot alter payment records or product pricing.
Product manager	Manage products, rules, translations and publication workflow.	Cannot access unnecessary claim medical documents.
Insurer administrator	Manage organization users, roles, reports and operational configuration.	Cannot delete audit records.
Platform administrator	Tenant configuration, technical support and system operations.	Customer-content access must be exceptional, justified and logged.
Auditor / reviewer	Read-only access to approved reports, logs and evidence.	No transactional updates.

10.2 Core status models

Object	Proposed statuses
Enrollment	DRAFT, SUBMITTED, PENDING_REVIEW, APPROVED, REJECTED, CANCELLED
Payment	CREATED, PENDING, SUCCESSFUL, FAILED, REVERSED, RECONCILIATION_REQUIRED
Policy	PENDING_PAYMENT, ACTIVE, GRACE_PERIOD, LAPSED, EXPIRED, CANCELLED
Claim	DRAFT, SUBMITTED, UNDER_REVIEW, INFORMATION_REQUIRED, APPROVED, REJECTED, PAYOUT_PENDING, PAID, APPEALED, CLOSED
Complaint	OPEN, ASSIGNED, INVESTIGATING, RESOLVED, ESCALATED, CLOSED

10.3 Representative REST API surface

Method and endpoint	Purpose
POST /api/v1/auth/register	Create a citizen account and verification challenge.
POST /api/v1/auth/login	Authenticate and issue secure access and refresh tokens.
GET /api/v1/products	List published products available to the authenticated user.
GET /api/v1/products/{id}	Return product benefits, exclusions, pricing and documents.
POST /api/v1/enrollments	Create an enrollment application.
POST /api/v1/enrollments/{id}/consent	Record versioned consent and policy acceptance.
POST /api/v1/payments/sandbox	Create a sandbox premium payment.
GET /api/v1/policies	List the customer's policies.
GET /api/v1/policies/{id}/card	Return digital policy-card data.
GET /api/v1/verify/{token}	Return limited, non-sensitive policy verification status.
POST /api/v1/claims	Create a claim draft.
POST /api/v1/claims/{id}/documents	Upload a claim document using controlled object storage.
POST /api/v1/claims/{id}/submit	Validate and submit a claim.
PATCH /api/v1/admin/claims/{id}/status	Authorized insurer status transition with reason and audit event.
POST /api/v1/claims/{id}/appeals	Submit an appeal against a claim decision.
GET /api/v1/admin/reports/overview	Return insurer operations and impact indicators.

API quality requirements

Use OpenAPI documentation, consistent error bodies, idempotency keys for payments and callbacks, pagination for all lists, correlation IDs, input validation, optimistic locking where needed and backward-compatible versioning.

11. Enterprise Architecture and Spring Boot Design

11.1 Recommended architecture strategy

Build the hackathon system as a modular monolith using Spring Boot. Each domain module has its own API, application, domain and persistence layers, and modules communicate through explicit interfaces and domain events. This gives the team one deployable artifact, one transaction boundary and faster testing while preserving a path to microservices.

For an enterprise rollout, high-change or high-volume modules can be extracted gradually. Payments, notifications, claims, reporting and partner integration are the most likely first candidates because they have independent scaling and reliability requirements.

Stage	Architecture	Reason
Six-day hackathon MVP	Spring Boot modular monolith + PostgreSQL + object storage	Fast delivery, simple debugging, reliable transactions and one deployment.
Pilot with insurer partner	Modular monolith with Redis, message broker, API gateway and production observability	Adds resilience and integration capability without unnecessary distribution.
Scaled enterprise platform	Selective microservices by bounded context	Independent scaling, teams, releases and fault isolation when justified by real load.

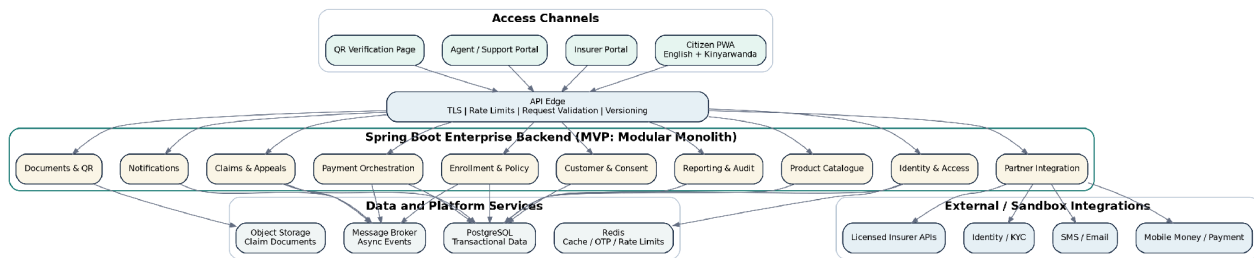


Figure 3. Enterprise target architecture with Spring Boot service boundaries

11.2 Proposed backend module structure

Module	Responsibility	Key domain objects
identity	Authentication, roles, permissions, sessions and privileged access.	User, Role, Permission, Session, MFChallenge
customer	Profile, language, occupation, consent and data rights.	CustomerProfile, ConsentRecord, DataRequest
partner	Insurer, branch, agent and agreement configuration.	Partner, PartnerUser, Agent, Agreement
product	Product catalogue, versions, benefits, exclusions and rules.	InsuranceProduct, ProductVersion, Benefit, Exclusion, PremiumPlan
enrollment	Application, eligibility, beneficiary and approval workflow.	Enrollment, EligibilityResult, Beneficiary
policy	Policy issuance, status, coverage period, renewal and QR token.	Policy, PolicyEvent, VerificationToken
payment	Premium request, callback, receipt and reconciliation.	Payment, PaymentAttempt, Receipt, ReconciliationItem
claim	Claim intake, evidence, review, decisions, payouts and appeals.	Claim, ClaimItem, Decision, Appeal, ClaimEvent

Module	Responsibility	Key domain objects
document	File metadata, secure object references, retention and access.	Document, DocumentAccessEvent
notification	Templates, in-app messages, SMS/email adapters and delivery.	Notification, Template, DeliveryAttempt
reporting	Operational dashboards, exports and scheduled aggregates.	ReportDefinition, MetricSnapshot
audit	Business audit, personal-data access logging and immutable events.	AuditEvent, DataAccessLog
integration	External API adapters, webhooks, credentials and retries.	IntegrationConfig, WebhookEvent, OutboxEvent
shared	Common errors, IDs, money, time, localization and technical utilities.	Value objects and shared infrastructure only

11.3 Suggested technology stack

Layer	Recommended technology	Purpose
Citizen and partner UI	Next.js / React Progressive Web App	Responsive, installable, low-friction access on phones and desktops.
Backend	Java 21 and a supported Spring Boot release	Enterprise REST APIs, domain logic, validation, integration and operations.
Security	Spring Security, OAuth2/OIDC-ready JWT, MFA for privileged users	Authentication, authorization and protection against common application attacks.
Persistence	PostgreSQL + Spring Data JPA + Flyway	Reliable transactions, relational integrity and versioned schema migration.
Caching and short-lived state	Redis	Rate limits, OTP state, short-lived tokens and cache where justified.
Documents	S3-compatible object storage such as MinIO for development	Secure claim evidence and policy documents separated from transactional rows.
Messaging	RabbitMQ or managed equivalent	Asynchronous notifications, integration retries and event-driven processing.
API documentation	Springdoc OpenAPI / Swagger UI	Interactive and versioned API documentation.
Testing	JUnit, Mockito, Testcontainers, REST Assured and Playwright	Unit, integration, API and end-to-end test coverage.
Deployment	Docker, CI/CD, managed container platform; Kubernetes only when needed	Repeatable releases and scalable operations.
Observability	Actuator, Micrometer, Prometheus, Grafana and OpenTelemetry	Health, metrics, logs, traces and alerting.

11.4 Enterprise engineering standards

- Use domain-driven package boundaries rather than placing all controllers, services and repositories in global folders.
- Keep controllers thin; business rules belong in application and domain services.
- Use database migrations from the first commit and never edit production schemas manually.
- Use an outbox pattern for reliable publication of important domain events when messaging is introduced.
- Use idempotency for payment requests, webhooks, claim submission and partner callbacks.
- Store money as precise decimal values with explicit currency and never as floating-point numbers.
- Apply optimistic locking to frequently updated operational records such as claims and policies.
- Use structured logs with correlation IDs and exclude secrets, tokens and unnecessary personal data.

- Provide health, readiness and liveness checks, but restrict sensitive operational endpoints.
- Automate unit, integration, security and end-to-end tests in CI before deployment.

12. Data, Security, Privacy and Regulatory Readiness

12.1 Regulatory positioning

Ingoboka should be positioned as a technology and distribution platform working with licensed insurers and authorized partners. A live pilot must be reviewed with the appropriate insurer compliance team and relevant authorities. The National Bank of Rwanda operates a regulatory sandbox framework for innovative financial services, which may provide an appropriate route for controlled testing depending on the final business model and regulatory classification. [3]

Rwanda's personal data protection and privacy law applies to automated and non-automated processing of personal data in Rwanda and to processing concerning data subjects located in Rwanda. The law defines health and medical information as sensitive personal data and requires appropriate processing safeguards. It also requires valid, informed consent where consent is the basis of processing, logging of personal-data operations and records of processing activities. [4][5]

12.2 Security controls

Control area	Required approach
Identity	Strong password policy, verified contact, short-lived access token, refresh-token rotation, MFA for insurer and platform administrators.
Authorization	Role-based and tenant-aware access, ownership checks, least privilege and method-level authorization.
Transport and storage	TLS, encrypted storage, secret management, database backups and encrypted object storage.
Sensitive documents	Private buckets, signed short-lived URLs, malware scanning hook, access logging and separate retention rules.
QR verification	Random opaque token that reveals only minimum status data; never encode national ID, medical or claim details.
Auditability	Append-only business audit and personal-data access log recording actor, purpose, time, action and result.
Application security	Input validation, secure headers, CSRF protection where relevant, rate limiting, dependency scanning and secure error handling.
Operational security	Centralized logs, alerting, backup verification, incident-response process and privileged activity review.

12.3 Privacy-by-design requirements

- Present privacy and product consent in a language the user understands and keep the exact version accepted.
- Collect only information necessary for the selected product and claim type.
- Separate customer profile data, claim documents and analytics views by purpose and authorization.
- Provide access, correction, withdrawal, deletion or restriction workflows where legally applicable.
- Define retention periods by data category and insurer obligation; do not keep documents indefinitely by default.
- Complete a data-protection impact assessment before processing sensitive health or large-scale customer data.
- Use formal data-processing agreements with cloud, messaging, identity and insurer partners.
- Review authorization requirements before storing or transferring personal data outside Rwanda.
- Never use customer data to train external AI models without an approved lawful basis and explicit governance.

12.4 Responsible use of AI

AI is optional and should not be required for the core MVP. If included, it may assist with document classification, translation drafts, missing-document detection or simplification of approved product text. It must not invent product terms, decide eligibility, set premiums or approve or reject claims. All generated text should be checked against the approved product version and important decisions must be made or confirmed by authorized humans.

13. Impact Framework and Success Measures

13.1 Expected citizen impact

- Greater awareness of relevant insurance products through understandable Kinyarwanda and English explanations.
- Lower enrollment friction through mobile-first and assisted journeys.
- Better fit for irregular incomes through insurer-configured contribution frequencies.
- Reduced loss of proof through a secure digital policy wallet.
- Higher trust through visible claim requirements, status updates, decision reasons and appeal options.
- Stronger household and micro-business resilience when insured events occur.

13.2 Expected insurer and ecosystem impact

- Lower marginal distribution and servicing cost for inclusive products.
- Structured data for product improvement, service quality and inclusion monitoring.
- Faster document collection, claim triage and customer communication.
- Improved auditability, reconciliation and partner integration.
- A reusable digital channel for cooperatives, associations, employers and community-based distribution.

13.3 Proposed measurable indicators

Impact area	Indicator	Initial pilot target type
Reach	Registered users, completed enrollments, active policies and districts represented.	Baseline and monthly growth
Inclusion	Share of women, youth, rural users, informal workers and Kinyarwanda-first users.	Representation, not token registration
Understanding	Users correctly answering key coverage and exclusion checks before consent.	Comprehension rate
Conversion	Needs assessment to application, application to payment and payment to active policy.	Funnel completion
Affordability	Chosen contribution frequency, failed-payment rate and lapse reasons.	Usage and failure pattern
Claims service	Submission completeness, time to first response, time to decision and reopened claims.	Service-level performance
Trust	Complaint rate, appeal rate, decision-explanation satisfaction and renewal intention.	Quality and fairness
Operations	Manual touches per policy/claim, integration failures and reconciliation exceptions.	Efficiency
Security and privacy	Access violations, consent evidence coverage, security incidents and data-rights response time.	Zero tolerance for unauthorized access

Impact principle

Success is not the number of downloads. Success is meaningful, informed use: users understand the cover, maintain an active policy and can complete a fair claims process.

14. Why This Is a Strong Hackathon Solution

- 1. Direct thematic alignment:** The official hackathon includes digital insurance as a primary area, while Ingoboka directly addresses access, distribution, customer understanding and claims.
- 2. Evidence-based national relevance:** The financial inclusion roadmap identifies low and uneven insurance penetration and specifically calls for affordable products, digital distribution and fintech-enabled claims.
- 3. Complete and demonstrable MVP:** Judges can see a real journey from registration to digital policy and claim decision, not only a dashboard or a predictive model.
- 4. Localized inclusion:** Kinyarwanda-first language, low-data PWA design, assisted mode and irregular contribution options are built around Rwandan citizens rather than copied from a foreign insurance app.
- 5. Responsible fintech design:** The concept preserves insurer authority, transparent consent, privacy, human claim decisions and appeal mechanisms.
- 6. Enterprise credibility:** Spring Boot, modular architecture, audit, versioning, secure documents, APIs and observability provide a believable path beyond the hackathon.
- 7. Partner value:** The platform gives insurers, cooperatives, agents and payment partners a shared operating layer and measurable channel.
- 8. Scalable scope:** The MVP can start with personal accident cover and later support health-related benefits, funeral, business assets and agriculture without rebuilding the platform.

14.1 Differentiation from a basic insurance app

Basic app approach	Ingoboka enterprise approach
Static list of products	Versioned, partner-owned product catalogue with rules, benefits, exclusions and localized explanations.
One payment button	Payment orchestration, callbacks, receipts, idempotency and reconciliation.
PDF policy download	Policy lifecycle, digital wallet, status events and minimum-data QR verification.
Claim upload form	Dynamic checklist, secure documents, officer workflow, timeline, decision reason and appeal.
Single administrator	Tenant-aware roles for insurers, product teams, claims officers, agents, support and auditors.
Attractive screens only	Working end-to-end APIs, data model, security, audit, tests and deployment.

15. MVP Scope and Six-Day Delivery Plan

15.1 Definition of done for the hackathon MVP

MVP acceptance journey

A citizen registers -> gives consent -> completes needs assessment -> selects a demo product -> completes sandbox payment -> receives an active digital policy and QR -> submits a claim with evidence -> insurer reviews it -> citizen sees a clear decision and can appeal.

15.2 Must-have features

- Responsive Kinyarwanda and English citizen interface.
- Citizen and insurer-user authentication with role-based authorization.
- One configurable demo insurance product with benefits, exclusions and contribution options.
- Needs assessment and product explanation.
- Enrollment, consent and beneficiary capture.
- Sandbox payment and receipt.
- Digital policy wallet and limited QR verification.
- Claim creation, secure document upload and claim timeline.
- Insurer claim review, information request, approval or rejection with reason.
- Basic reports, audit log, error handling, seed data and deployed demo.

15.3 Explicitly out of MVP

- Real underwriting, real premium collection or real claim payout.
- Direct integration with NBR, national ID, hospitals, eKash, MTN, Airtel or an insurer core system unless written access is obtained.
- Multiple production-grade insurance products.
- Autonomous AI decisions, fraud scoring or biometric identification.
- Microservices, Kubernetes and complex event streaming unless the core journey is already complete.
- Native Android and iOS applications; use a PWA for the hackathon.

15.4 Six-day execution plan

Day	Primary objective	Required output
1	Finalize product, architecture and design	Approved name, PRD, data model, Stitch screens, user flows, repository and deployment skeleton.
2	Build backend foundation	Spring Boot project, security, users, roles, PostgreSQL, migrations, OpenAPI and seed data.
3	Complete product-to-policy flow	Product catalogue, needs assessment, enrollment, consent, sandbox payment, policy activation and QR.
4	Complete claims flow	Claim forms, document upload, claim queue, review, decision reason, status history and appeal.
5	Polish enterprise qualities	Localization, validation, audit, notifications, reports, security checks, tests and error states.
6	Stabilize and submit	Production deployment, phone/laptop testing, demo accounts, README, video, pitch and backup recording.

15.5 Demo accounts and scenario

Account	Scenario
Citizen: Aline	Market trader enrolls in weekly personal accident cover, receives a policy card and submits a hospitalization claim.
Claims officer: Eric	Reviews Aline's claim, requests a missing document and later approves the sandbox decision.
Insurer manager: Diane	Views product uptake, payment success, claim queue and turnaround indicators.
Public verifier	Scans a QR and sees only policy reference, provider, product, active status and expiry.

16. Sustainability and Growth Model

16.1 B2B2C operating model

Ingoboka should serve citizens through licensed insurer partnerships rather than charging vulnerable users simply to access the platform. Revenue arrangements must be transparent, permitted by regulation and structured so that customer value is not reduced by hidden fees.

Potential revenue stream	Description	Condition
Partner subscription	Insurer pays for product, workflow, reporting and user administration.	Tiered by users, products or service level.
Implementation and integration	One-time setup, product configuration and core-system integration.	Clearly scoped professional service.
Transaction or administration fee	Fee associated with successful enrollment, payment or policy administration.	Only where legally and contractually allowed; fully disclosed.
Analytics and reporting	Aggregated operational dashboards and approved impact reporting.	No sale of personal data; use minimization and anonymization.
Cooperative or employer programs	Sponsored protection programs for members or workers.	Partner contract and customer consent.

16.2 Growth roadmap

Phase	Focus
Hackathon	Functional sandbox MVP and validated citizen/insurer journey.
Pilot	One licensed insurer, one product, one target community or cooperative, controlled transactions and support.
Multi-partner platform	Multiple insurers, product marketplace, payment reconciliation, partner APIs and stronger reporting.
National scale	Broader district coverage, agent networks, formal SLAs, high availability, advanced risk controls and regulator-ready reporting.
Regional potential	Configurable country, currency, language and regulatory modules after Rwanda product-market fit.

17. Risks, Assumptions and Boundaries

Risk	Potential effect	Mitigation
Regulatory classification	The service may be treated as an intermediary or regulated financial service.	Engage licensed insurers and NBR early; use sandbox pathway where appropriate; avoid unlicensed activities.
No insurer partnership during hackathon	Product terms and claim decisions are not real.	Use clearly labeled demo products and sandbox workflows; never present them as approved commercial cover.
Sensitive customer and claim data	Privacy harm, legal exposure and loss of trust.	Minimize data, encrypt, restrict, log, define retention and complete DPIA before pilot.
Payment integration failure	Policy activation or reconciliation errors.	Use idempotent callbacks, status reconciliation, retries and manual exception queue.
Fraud or false claims	Financial loss and unfair delay for legitimate users.	Evidence checklist, duplicate signals, manual review and future partner fraud tools.
Low comprehension	Users enroll without understanding.	Plain language, visual explanations, comprehension checks, Kinyarwanda and assisted support.
Scope creep in six days	Incomplete core journey and unstable demo.	Freeze must-have scope; defer AI, microservices and extra products.
AI-generated misinformation	Incorrect product explanation or unfair decision.	Make AI optional, ground outputs in approved text and prohibit automated final decisions.
Cloud or external transfer obligations	Non-compliant storage or processing.	Review Rwanda DPP requirements, partner contracts and required authorizations before production.

17.1 Key assumptions

- The hackathon permits sandbox or simulated financial transactions.
- The platform will demonstrate insurer-side decisions using authorized demo users.
- A responsive PWA is acceptable as the primary citizen channel.
- Kinyarwanda content will be reviewed by native speakers and product/legal experts.
- All commercial insurance products will be supplied and approved by licensed partners.
- Final hosting, data location and integration choices will be reviewed during pilot planning.

18. Conclusion

Ingoboka is a strong response to the digital-insurance challenge because it solves a practical service problem rather than presenting technology without a clear user need. It combines localized access, product understanding, flexible contribution support, a secure policy wallet and transparent claims in one end-to-end journey.

The concept is ambitious enough to show enterprise potential but can still be demonstrated credibly within six days when the team focuses on one product, sandbox integrations and the full customer-to-insurer workflow. Spring Boot provides a suitable foundation for secure APIs, transactional consistency, modular domain design and long-term integration.

The most important success factor is disciplined positioning: Ingoboka is a technology platform that helps licensed insurers serve citizens better. A compelling prototype should make that partnership model, responsible design and measurable citizen impact visible in every screen and every workflow.

Appendix A - Form-Ready Problem Description (Maximum 250 Words)

Problem description

Although Rwanda has achieved strong overall financial inclusion, meaningful access to insurance remains limited and uneven, especially among informal workers, young people, women and citizens outside major urban areas. Market vendors, moto riders, farmers, casual workers and micro-entrepreneurs often earn irregular incomes that do not fit conventional insurance payment schedules. Many potential customers also struggle to compare products, understand benefits and exclusions, keep proof of cover and identify the documents required when making a claim.

The problem is not only the availability of insurance products. It is the absence of a simple, understandable and transparent journey connecting citizens to suitable products from licensed insurers. Enrollment, payment records, policy documents, claims evidence and customer communication may be fragmented. After an insured event, a customer may not know what to submit, what is missing, who is reviewing the claim or why a decision was made.

As a result, economically active but underinsured citizens remain vulnerable to accidents, hospitalization, loss of income, funeral costs and damage to productive assets. They may be forced to consume savings, borrow, sell assets or depend on relatives. Insurers also face high costs when serving low-premium customers through manual processes. A localized digital platform is needed to make approved insurance easier to understand, access, manage and claim while preserving insurer authority, data privacy and responsible customer protection.

Approximate word count: 219

Appendix B - Form-Ready Solution Description

Solution description

Ingoboka is a multilingual digital microinsurance, policy-wallet and claims platform designed for Rwandan citizens with irregular incomes. It connects citizens to approved products from licensed insurers through a mobile-first, low-data experience. Users can complete a short needs assessment, understand benefits and exclusions in simple Kinyarwanda or English, select an insurer-configured contribution schedule, complete a sandbox payment, receive a secure digital policy card and submit a claim with supporting documents.

The insurer portal enables authorized teams to configure products, monitor enrollment, review claims, request missing information, record reasoned decisions and view operational reports. Every important action is recorded in an audit trail. The platform does not underwrite risk or replace insurers; it provides the digital distribution and service infrastructure that helps approved providers reach and serve underserved customers more efficiently.

Appendix C - 60-Second Pitch

Pitch script

Millions of Rwandans work, trade and earn every day, but an accident or hospitalization can still wipe out their savings because insurance is difficult to understand, pay for and claim. Ingoboka changes that.

Ingoboka is a Kinyarwanda-first digital platform connecting citizens to approved microinsurance from licensed insurers. A user can understand the cover, choose a contribution schedule that fits irregular income, receive a digital policy card and follow a claim from submission to decision. Insurers receive a secure portal for product configuration, claim review, reporting and audit.

We are not replacing insurers. We are giving them an enterprise digital channel to reach underserved citizens responsibly. Our Spring Boot MVP demonstrates the complete journey from enrollment to claim resolution, with privacy, transparency and human decision-making built in. Ingoboka means protection that is simple, understandable and accessible to everyone.

Appendix D - High-Level Data Model

Entity	Purpose	Important relationships
User / Role / Permission	Authentication and authorization.	User has roles; roles have permissions; user belongs to a partner tenant or is a citizen.
CustomerProfile / ConsentRecord	Citizen data, preferences and consent evidence.	One profile per citizen; many versioned consents.
Partner / PartnerUser / Agent	Insurer and channel organizations.	Partner owns products and users; agents are assigned to partner or program.
InsuranceProduct / ProductVersion	Approved product configuration and history.	Product has many immutable versions, benefits, exclusions and premium plans.
Enrollment / Beneficiary	Customer application and selected coverage.	Enrollment links customer, product version and beneficiary.
Policy / PolicyEvent / VerificationToken	Issued contract record and lifecycle.	Policy originates from approved enrollment and has events and QR tokens.
Payment / PaymentAttempt / Receipt	Premium transactions and reconciliation.	Payments link to enrollment or policy and may have multiple attempts.
Claim / ClaimDocument / ClaimEvent	Claim submission, evidence and status history.	Claim belongs to policy and has documents, assignments, decisions and events.
Appeal / Complaint	Customer challenge and support case.	Appeal links to a claim decision; complaint may link to any service object.
Notification / DeliveryAttempt	Customer and staff communication.	Notification uses versioned template and records channel attempts.
AuditEvent / DataAccessLog	Evidence of business and personal-data operations.	Append-only records reference actor, object, reason, time and result.
IntegrationConfig / WebhookEvent / OutboxEvent	Reliable external communication.	Tenant-scoped configuration and retryable event records.

Appendix E - References

- [1] National Bank of Rwanda, Fintech Innovation Hackathon 2026 registration form, pages 1-4, accessed 14-15 June 2026. The form identifies the digital-insurance thematic area, application stage and optional product video.
- [2] National Bank of Rwanda and Ministry of Finance and Economic Planning, National Financial Inclusion Roadmap 2026-2030. Relevant sections include insurance penetration, target of 15% insured adults, digital distribution, affordable microinsurance and fintech-enabled claims processing.
- [3] National Bank of Rwanda, Regulation No. 41/2022 governing the Regulatory Sandbox and NBR regulatory sandbox information page.
- [4] Rwanda Data Protection and Privacy Office, Law relating to the protection of personal data and privacy, Articles 1-3: purpose, scope and definitions.
- [5] Rwanda Data Protection and Privacy Office, Law relating to the protection of personal data and privacy, Articles 4-17: consent, sensitive-data safeguards, logging and records of processing.

[6] Spring Boot Reference Documentation, documentation overview and production guidance.

[7] Spring Security Reference Documentation, authentication, authorization and protection against common attacks.

Final disclaimer

This document is a product and technical concept, not legal, actuarial or regulatory advice. Production launch requires licensed insurer participation, approved product terms, security and privacy assessment, contractual agreements and regulatory confirmation.